

Brewed to Satisfaction: Analyzing Key Factors Influencing Student Satisfaction in Local Coffee Shops

Final Class Project Presentation
Group 3
December 4, 2024

By: Xiao Ran (Christine) Zhu, Neo La, Mary Dorothy Alinabon, Adrian Cardenas





Agenda



01

Introduction

02

Problem Statement

03

Objective

04

**Method/Analytical
Approach**

05

**Data Summary/Data
Visualization**

06

Results

07

**Conclusions and
recommendations**

08

Lessons Learned

09

Challenges



Introduction

- Coffee shop are vital to college students
 - Spaces for studying, socializing, relaxation, etc.
- Local coffee shops around SJSU
 - Student-tailored environment
- Project focus:
 - Analyze customer satisfaction
 - Ambiance
 - Drink quality
 - Price
 - Customer service



Problem Statement

Problem: Many local coffee shops around SJSU struggle to attract students, despite offering unique, student-focused environments.



Understanding why students underutilize these spaces is crucial to helping local businesses thrive and better meet student needs.

Objective

- To explore factors driving student preference
 - Comparing scores:
 - 7Leaves
 - Gongcha
 - Breaktime
 - Big name: Starbucks
- Help local coffee shops...
 - Address gaps
 - Compete more effectively
 - Attract more students
 - Foster community



Gong cha

Methodology



Analytical Approach

Method: ANOVA

Parameters:

- $\mu_1 = 7$ Leaves
- $\mu_2 = \text{Break Time}$
- $\mu_3 = \text{Gong Cha}$
- $\mu_4 = \text{Starbucks}$

Null Hypothesis

- $H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$

Alternative Hypothesis

- $H_a: \mu_1 \neq \mu_2 \neq \mu_3 \neq \mu_4$

Assumptions:

- Level of significance (α) = 0.05
- Independence: Each observation is independent of the others
- Normality: The customer satisfaction scores are approximately normally distributed
- Equal variances: The variance among the scores in different coffee shops is assumed to be equal

ANOVA Test Equation: $Y_i = \mu + \tau_i + \epsilon_i$

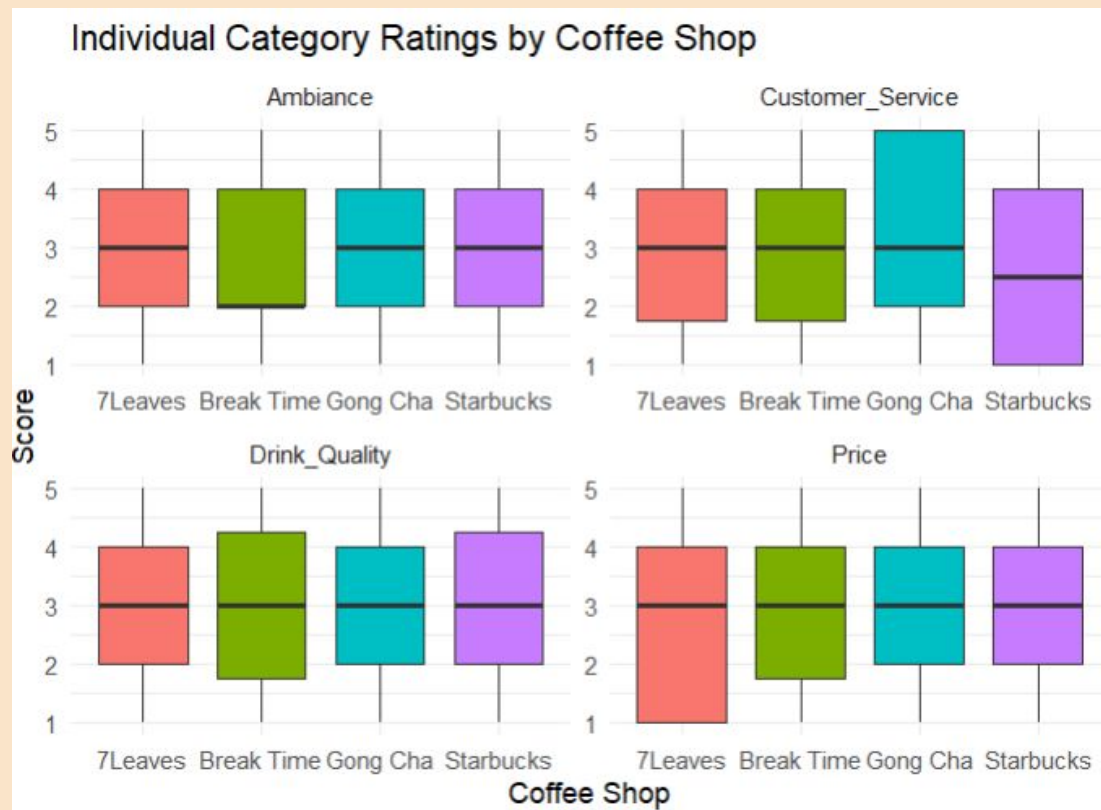
- Y_i : Observed satisfaction score.
- μ : Overall mean satisfaction score across all coffee shops.
- τ_i : The specific shop's effect (how it deviates from the average).
- ϵ_i : Random error (unpredictable, individual-specific factors).

Data Summary

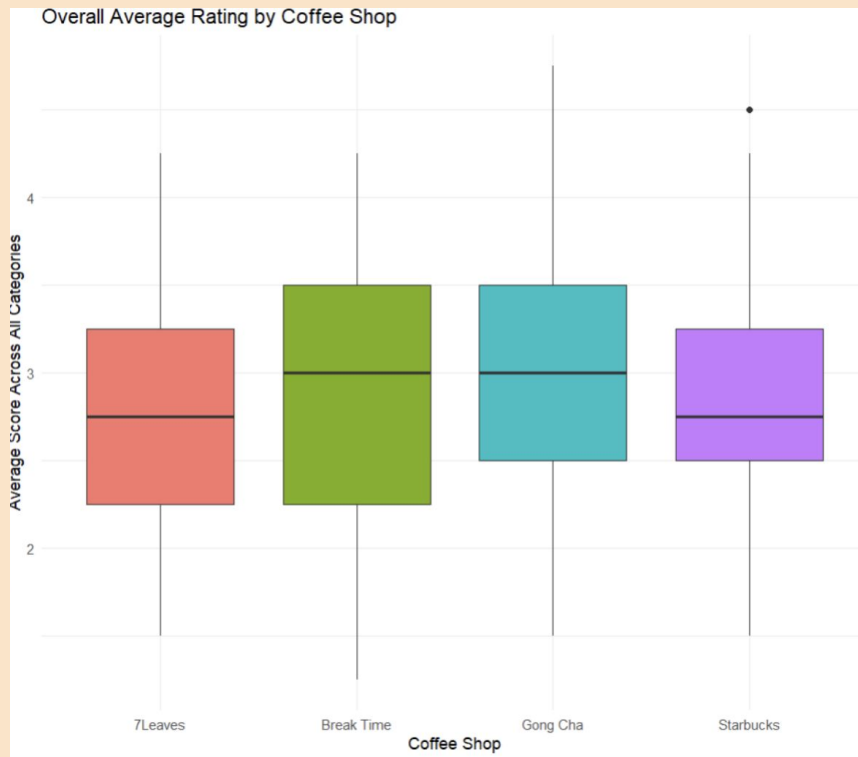
Coffee_Shop	Ambiance	Drink_Quality	Price	Customer_Service
Length:240	Min. :1.000	Min. :1.00	Min. :1.000	Min. :1.0
Class :character	1st Qu.:2.000	1st Qu.:2.00	1st Qu.:2.000	1st Qu.:2.0
Mode :character	Median :3.000	Median :3.00	Median :3.000	Median :3.0
	Mean :2.904	Mean :2.95	Mean :2.871	Mean :2.9
	3rd Qu.:4.000	3rd Qu.:4.00	3rd Qu.:4.000	3rd Qu.:4.0
	Max. :5.000	Max. :5.00	Max. :5.000	Max. :5.0

	Coffee_Shop	Ambiance	Drink_Quality	Price	Customer_Service
1	7Leaves	3.00	2.88	2.65	2.82
2	Break Time	2.85	2.98	2.82	2.82
3	Gong Cha	3.03	2.72	3.03	3.30
4	Starbucks	2.73	3.22	2.98	2.67

Data Visualization



Data Visualization



Independence

Ambiance vs Drink Quality :

Spearman rho: -0.087

p-value: 0.1767

Conclusion: Independent

Ambiance vs Price :

Spearman rho: -0.037

p-value: 0.5636

Conclusion: Independent

Ambiance vs Customer Service :

Spearman rho: 0.017

p-value: 0.7884

Conclusion: Independent

Drink Quality vs Price :

Spearman rho: -0.096

p-value: 0.1392

Conclusion: Independent

Drink Quality vs Customer Service :

Spearman rho: -0.027

p-value: 0.6789

Conclusion: Independent

Price vs Customer Service :

Spearman rho: 0.105

p-value: 0.1046

Conclusion: Independent

The Spearman correlation results show that all satisfaction categories are independent of one another.

Levene tests

Levene's Test for Price Rating:

Levene's Test for Homogeneity of Variance (center = median)

	Df	F value	Pr(>F)
group	3	0.4024	0.7514
	236		

Levene's Test for Ambiance Rating:

Levene's Test for Homogeneity of Variance (center = median)

	Df	F value	Pr(>F)
group	3	0.3778	0.7691
	236		

Levene's Test for Drink_Quality Rating:

Levene's Test for Homogeneity of Variance (center = median)

	Df	F value	Pr(>F)
group	3	0.9598	0.4124
	236		

Levene's Test for Customer_Service Rating:

Levene's Test for Homogeneity of Variance (center = median)

	Df	F value	Pr(>F)
group	3	0.1734	0.9143
	236		

Variances across all categories are equal, satisfying a key ANOVA assumption.

Equal variances ensure that differences in satisfaction means between coffee shops are not skewed by uneven variability in scores.

ANOVA can reliably test for significant differences in satisfaction across coffee shops.

RESULTS

- ANOVA Test Result:

```
ANOVA Results for Individual Categories:  
> for (category in names(anova_results)) {  
+   p_value <- anova_results[[category]]  
+   cat(category, ": p-value = ", round(p_value, 2), "\n")  
+ }  
Ambiance : p-value = 0.61  
Drink_Quality : p-value = 0.28  
Price : p-value = 0.43  
Customer_Service : p-value = 0.08  
> |
```

With a significance level of 0.05, all our ANOVA p-values are above 0.05. This means there are no big differences in satisfaction for ambiance, drink quality, price, or customer service between the coffee shops. The scores are similar, so these factors don't explain why students might like one shop more than another.



Conclusion and Recommendations



Coffee Shop Study Findings:

- **Similarities:** Ambiance, price, drink quality, customer service
- **Differentiators:** Proximity to campus, WiFi quality, Safety/Location

Recommendations:

- **Loyalty Programs:** Encourage repeat visits through student loyalty incentives.
- **University Partnerships:** Collaborate with university and student clubs for increased visibility.
- **Student Ambassadors:** Employ students to promote the shop and attract peers.



Lessons Learned

- Understood R better, explored more through different plotting techniques, and learned features for data analysis and visualization.
- Pairwise Comparison is not needed when the ANOVA test when there is no significant differences in the ANOVA test
- Learned how to set up timed meetings with the groups before important deadlines like using when2meet



Fin